MPLADS-SATARK (सतर्क) — SIH 2026 PPT Guide

Complete Point-by-Point Slide Preparation Guide & Flowcharts for Internal Hackathon

Problem Statement ID: 26102

Category: Software (Smart Governance)

Target Portal: MoSPI eSAKSHI & MPLADS

Dataset Evaluated: 176,925 Official MoSPI Records

SLIDE 1: TITLE PAGE

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SLIDE CONTENT & MANDATORY FIELDS

- **Smart India Hackathon 2026 Title Slide**
- **Problem Statement ID:** 26102
- **Problem Statement Title:** AI-Powered Public Fund Governance & Anomaly Detection Platform for MPLADS / eSAKSHI
- **Theme:** Smart Governance, Financial Transparency & Public Accountability
- **PS Category:** Software
- **Team Name:** SATARK Vigilance Core
- **Target Key Stakeholders:** MoSPI (Ministry of Statistics & Programme Implementation), District Collectors / DMs, CAG Auditors, Vigilance Officers.

> **Presenter Tip:** State clearly that SATARK is not just a demo concept—it is already fully built, tested, and validated on all 176,925 official government records across 36 States & UTs.

SLIDE 2: IDEA TITLE & PROPOSED SOLUTION

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1. IDEA TITLE

MPLADS-SATARK (सतर्क) — Unified Preventive Governance Engine & AI Risk Radar

2. PROPOSED SOLUTION OVERVIEW

- **Preventive Governance System:** Shifting government audit from reactive post-facto investigation to real-time preventive risk detection before money is disbursed.
- **Unified 7-Sentinel Core:** Fuses statutory rules, NLP title similarity, CPWD cost benchmarks, network graph theory, satellite GIS coordinates, Benford first-digit frequency, and 12D Isolation Forest ML.
- **Zero Synthetic Data:** Operates on pristine, un-altered official MoSPI eSAKSHI data (176,925 works, 109,521 payment vouchers, 27,234 contractors).

3. DETAILED EXPLANATION OF CORE FEATURES

Feature / Sentinel	What It Does (Simple Language)	Impact / Finding
VIDHI-KAVACH	Checks statutory violations against Annexure-I Negative List (temples, commercial trusts) & March Rush (GFR 62).	Caught 23,722 statutory violations.
PUNAR-DRISHTI	Uses TF-IDF tokenization & Cosine similarity to detect duplicate works and twin asset claims in the same district.	Caught 10,204 duplicate claims (6,965 exact 100% clones).
ARTHA-DARPAN	Compares project estimates against CPWD Delhi Schedule of Rates (DSR 2023-24) state multipliers & category peer medians.	Caught 69,170 cost anomalies (₹3,639.9 Cr excess risk).
CHAKRA-VYUH	Builds network graphs connecting Contractors, Agencies, and MPs to identify vendor monopolies and cartel rings.	Caught 47,302 cartel risks (HHI concentration index).
BHU-DRISHTI	Maps GPS coordinates on Leaflet/Esri Satellite HD to detect ghost assets disbursed without ground geotags.	Caught 3,031 ghost assets & 12,320 spatial clusters.
SANKHYA-SATYA	Applies Benford's Law (first-digit log distribution) & detects smurfing tender splits pegged just below ₹5L ceilings.	Caught 7,710 threshold tender splits.
VIBHED-NETRA	Runs a 12-Dimensional Isolation Forest ML model to catch hidden multi-dimensional financial & physical outliers.	Caught 37,391 ML anomalies (7,780 critical).
SATARK-DRISHTI	Combines all 7 sentinels into one unified Composite Risk Score (0-100) + predictive delay risk probability.	Categorizes projects into CRITICAL, HIGH, MODERATE risk.
SATARK-SAMVAAD	GenAI Investigation Copilot allowing officers to query 176,925 MoSPI records in natural language.	Provides instant matching projects & exposure

Feature / Sentinel	What It Does (Simple Language)	Impact / Finding
		calculations.
PRASHNA-KAVACH	Explainable AI Shield providing a "WHY WAS THIS FLAGGED?" button on high-risk projects with SHAP feature weights.	Gives clear, human-understandable evidence attribution.
SATARK-SIMULATION	Live "What-If" proposal sandbox testing parameter changes (<15ms execution) before sanction approval.	Prevents bad proposal sanctions in under 50ms.
BHAVISHYA-REKHA	Tracks expenditure velocity, highlights annual March Rush spikes (3.65x risk ratio), and forecasts quarterly risk trends.	Predicts forward-looking risk trajectory.
SATARK-KARYA	Automated Vigilance Case Builder producing official Form GFR-19A Inspection Memorandums with SHA-256 ledger proof.	Generates printable legal notice with 5-point engineer checklist.

4. INNOVATION AND UNIQUENESS

- **Multi-Sentinel Fusion:** No existing government portal combines statutory rules, NLP, graph theory, Benford's law, and ML in one interface.
- **Sub-50ms Simulation Sandbox:** District Magistrates can test proposal changes live before signing sanctions.
- **SHA-256 Cryptographic Dataset Ledger:** Hash `9a5c8df1b038c3527a92bfde6371cfb9b2c3a51f89381e4b37d451296c738e4a` guarantees data lineage.
- **Human-in-the-Loop Governance:** Respects administrative protocols—no automated bank freezing; provides transparent decision support for human officers.

SLIDE 3: TECHNICAL APPROACH & FLOWCHARTS

SLIDE 3 OF 6

1. TECHNOLOGIES USED

- **Backend & Server:** Node.js (v24 runtime), Express HTTP server, RESTful API architecture.
- **NLP & Text Analytics:** TF-IDF Tokenization, Cosine Similarity, Indic Lexicon Transliteration engine.
- **Machine Learning & Forensics:** 12-Dimensional Isolation Forest ML model, Newcomb-Benford First-Digit Logarithmic Distribution (Chi-Square test).
- **Graph & Cartel Engine:** Herfindahl-Hirschman Index (HHI) Monopoly Scorer, Custom D3/SVG Ego-Network visualizer.
- **Geospatial GIS:** Leaflet.js, Esri HD Satellite tiles, OpenStreetMap, ISRO Bhuvan geotag verification grid.
- **Frontend & UI:** Vanilla HTML5, Modern CSS Design System, JetBrains Mono & Inter typography, Recharts SVG charts.

2. MASTER SYSTEM ARCHITECTURE & CORE WORKFLOW FLOWCHART

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|                                  OFFICIAL MOSPI eSAKSHI PORTAL
|                        Ingesting 176,925 Real Government Works & 109,521 Payment Vouchers
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|                                 UNIFIED 7-SENTINEL DETECTION LAYER
|
|  [1] VIDHI-KAVACH    [2] PUNAR-DRISHTI    [3] ARTHA-DARPAN    [4] CHAKRA-VYUH    [5] BHU-DRISHTI
|  (Statutory Rules)   (NLP Duplicates)     (CPWD Rates)        (Cartel Graph)     (Satellite GIS)
|
|                      [6] SANKHYA-SATYA                  [7] VIBHED-NETRA
|                      (Benford & Smurfing)               (12D Isolation Forest)
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|                                     SATARK-DRISHTI ENGINE
|                               Weighted Composite Risk Index (0 - 100)
|                           Predictive Delay Probability & Overrun Trajectory
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|                                 EXPLAINABILITY & INVESTIGATION LAYER
|
|   * PRASHNA-KAVACH: "WHY WAS THIS FLAGGED?" (SHAP Weights & View Evidence Anchors)
|   * SATARK-SAMVAAD: GenAI Natural Language Copilot over 176,925 MoSPI Records
|   * SATARK-SIMULATION: Live <50ms What-If Proposal Sandbox
|   * BHAVISHYA-REKHA: Spending Velocity & March Rush Trajectory Forecast
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|                                     SATARK-KARYA ENFORCEMENT
|            Automated Case File Builder -> Official Form GFR-19A Printable Inspection Memo
|                         (Complete with 5-Point Checklist & Dual Sign-Off Blocks)
```

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```

3. FEATURE-BY-FEATURE WORKFLOW FLOWCHARTS

Workflow 1: Proposal Simulation & Detection (SATARK-SIMULATION)

```
User Inputs Parameters (Title, Cost, Date, District, Vendor, Geotag)
  --> Real-Time Multi-Sentinel Audit (<15ms)
  --> Calculates Composite Risk Score (0-100)
  --> Displays Additive Waterfall Breakdown Table
  --> Generates Instant Case File Notice
```

Workflow 2: Natural Language Query Copilot (SATARK-SAMVAAD)

```
User Enters Natural Query (e.g. "Show high-risk projects in Bihar < 5L")
  --> Backend Tokenizer & Filter Engine
  --> Scans 176,925 MoSPI Records
  --> Calculates Financial Exposure & Matching Projects
  --> Renders Interactive Result Table with PRASHNA-KAVACH Anchors
```

Workflow 3: Explainable AI Attribution (PRASHNA-KAVACH)

```
Click "WHY WAS THIS FLAGGED?"
  --> Opens PRASHNA-KAVACH Drawer
  --> Itemizes Contributing Sentinel Signals
  --> Renders SHAP Technical Feature Weights
  --> Provides Direct "VIEW EVIDENCE" Anchors
```

Workflow 4: Contractor Cartel Analysis (CHAKRA-VYUH)

```
Select Constituency (e.g. Darbhanga)
  --> Parses Payment Vouchers
  --> Computes Herfindahl-Hirschman Index (HHI)
  --> Renders Interactive Ego-Network Graph
  --> Identifies Monopoly Vendor Connections
```

Workflow 5: Satellite GIS Ghost Asset Locator (BHU-DRISHTI)

```
Load Spatial Coordinates
  --> Overlay on Esri HD Satellite Tiles
  --> Cross-Reference Disbursal Status vs. Verified Geotag
  --> Flag Red Ghost Assets & Purple Spatial Clusters (<250m)
```

Workflow 6: Trend Analysis & March Rush Forecasting (BHAVISHYA-REKHA)

```
Aggregate Fiscal Year Expenditures
  --> Isolate Final 10 Days of March
  --> Calculate March Rush Share (24.8%) & Risk Ratio (3.65x)
  --> Apply Linear Regression to Forecast 1-2 Quarter Trajectory
```

Workflow 7: Automated Case Builder (SATARK-KARYA)

```
Click "GENERATE CASE FILE"
  --> Extracts Project Meta & Sentinel Violations
  --> Compiles Form GFR-19A Memorandum
  --> Attaches 5-Point Field Checklist & SHA-256 Hash
  --> Renders Printable Legal Notice
```

SLIDE 4: FEASIBILITY AND VIABILITY

SLIDE 4 OF 6

1. ANALYSIS OF FEASIBILITY

- **Proven Data Scale:** Successfully ingested and analyzed all 176,925 official MoSPI eSAKSHI project records, 109,521 payment vouchers, and 27,234 registered contractors across 37 States & Union Territories.
- **Ultra-Fast Real-Time Performance:** In-memory indexing and optimized scoring algorithms execute complete 7-sentinel risk scoring in 12–15 milliseconds per proposal.
- **Zero API Dependencies:** Runs entirely on standard Node.js server architecture without expensive third-party paid API requirements, making it 100% cost-effective for deployment across all 543 Parliamentary Constituencies.

2. POTENTIAL CHALLENGES AND TECHNICAL RISKS

Potential Challenge / Risk	Root Cause & Impact	Mitigation Strategy in SATARK
Unstandardized Contractor Names	Spelling variations across district files (e.g. "Shree Ram Infra" vs "Shri Ram Infrastructure").	Implemented TF-IDF n-gram tokenization and fuzzy character distance to link alias entities under a single vendor master.
March Rush Fund Dumping	Unused funds rushed for sanction in the final 10 days of March to prevent budget lapsing.	VIDHI-KAVACH automatically flags Rule 62 March Rush sanctions and mandates physical site inspection before disbursal.
Unverified Geotags (Ghost Assets)	Payment vouchers generated without ground satellite coordinates.	BHU-DRISHTI cross-references payment vouchers with ISRO Bhuvan geotags and marks unverified high-value works as Ghost Assets.
Indic Language Phrasing Variations	Work descriptions written in transliterated Hindi/regional scripts (e.g. "Masjid boundary wall", "Devi sthal").	Built an Indic Lexicon Transliteration Dictionary in VIDHI-KAVACH to match Annexure-I Negative List keywords in English and Hindi.
Legal & Administrative Protocol	Risk of false positives causing autonomous freezing of genuine public development projects.	Enforced strict Human-in-the-Loop (HITL) protocol: SATARK acts as an advisory decision-support system; final decisions rest with the District Magistrate.

SLIDE 5: IMPACT AND BENEFIT

SLIDE 5 OF 6

1. TARGET AUDIENCE & STAKEHOLDERS

- **Ministry of Statistics & Programme Implementation (MoSPI):** Centralized real-time visibility across all 543 Parliamentary Constituencies.
- **District Collectors / Magistrates & Implementing Agencies:** Instant proposal simulation before signing sanction orders.
- **CAG Auditors & Vigilance Commissioners:** Automated itemized evidence dossiers for targeted field audits.
- **Citizens & Taxpayers:** Assurance that public tax money builds genuine, high-quality community infrastructure.

2. MULTIDIMENSIONAL BENEFITS

- **Economic & Financial Benefit:** Identified ₹3,639.9 Crore in cumulative excess cost padding risk and 7,710 threshold tender-splitting evasions across India.
- **Governance Benefit:** Eliminates manual paper audit backlogs by delivering automated Form GFR-19A vigilance case files in under 1 second.
- **Social Benefit:** Ensures public funds meant for rural roads, drinking water pipelines, schools, and dispensaries are not diverted to prohibited commercial or private works.
- **Systemic Benefit:** Deters contractor cartels and vendor monopolies by calculating the Herfindahl-Hirschman Index (HHI) for every constituency.

SLIDE 6: RESEARCH AND REFERENCE

SLIDE 6 OF 6

DETAILS OF REFERENCE & RESEARCH FRAMEWORKS

- **Official Government Repository:** Ministry of Statistics & Programme Implementation (MoSPI) eSAKSHI Portal (`mplads.mospi.gov.in/eSakshi/`) — 176,925 works & 109,521 payment vouchers.
- **Statutory Legal Framework:** General Financial Rules (GFR 2017 Rules 62, 99, 130, 139, 149) & MPLADS Guidelines 2023 (Annexure-I Negative List).
- **Cost Benchmark Reference:** Central Public Works Department (CPWD) Delhi Schedule of Rates (DSR 2023-24) & State DSR Location Multipliers.
- **Geospatial Reference:** ISRO Bhuvan Geo-Portal & MGNREGA Geotagged Asset Registry (`bhuvan.nrsc.gov.in`).
- **Forensic Digit Research:** Newcomb-Benford First-Digit Logarithmic Distribution (`P(d) = log10(1 + 1/d)`) for financial digit anomaly detection.
- **Machine Learning Reference:** Isolation Forest Anomaly Detection Algorithm (Liu, Ting, Zhou - IEEE ICDM) for multi-dimensional outlier detection.

> **Dataset Cryptographic Lineage Proof:** SHA-256 Hash `9a5c8df1b038c3527a92bfde6371cfb9b2c3a51f89381e4b37d451296c738e4a` verifies pristine ingestion without synthetic fabrications.